Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 1

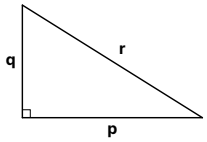
Scan to practise this topic online

KEY WORDS right-angled triangle *(tapatoru hāngai)* · hypotenuse *(tapa roa)* · square *(tapawhā rite)*

A The hypotenuse ●○○

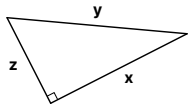
1. Write the letter of the hypotenuse of each triangle.

a)



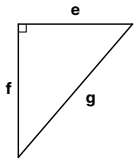
Hypotenuse: _____

b)



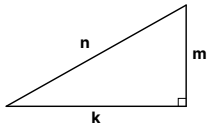
Hypotenuse: _____

c)



Hypotenuse: _____

d)



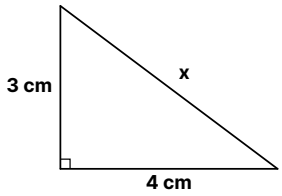
Hypotenuse: _____

B Finding the hypotenuse ●○○

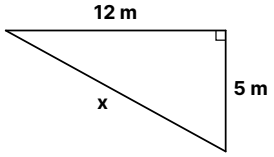
Remember: $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side.

2. Find the hypotenuse, x. Round to 1 decimal place where needed. Diagrams are not to scale.

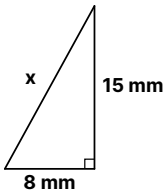
a)



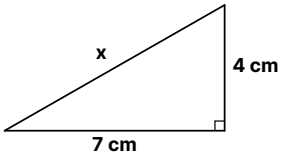
b)



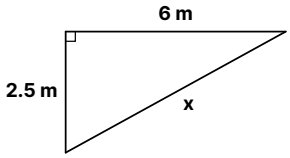
c)



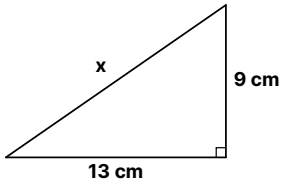
d)



e)



f)



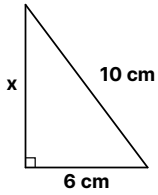
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C Finding a shorter side ●●○

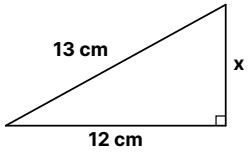
⚠ **Watch out:** Here the hypotenuse is given, so subtract: $a^2 = c^2 - b^2$.

3. Find the length of the side marked x. Round to 1 decimal place where needed.

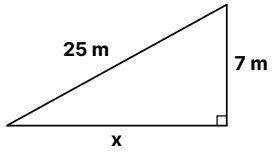
a)



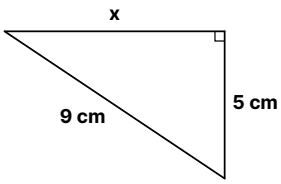
b)



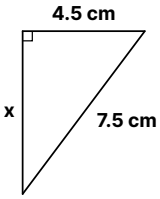
c)



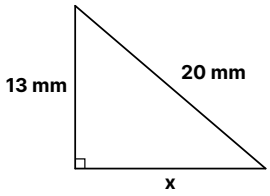
d)



e)



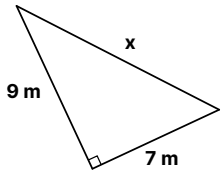
f)



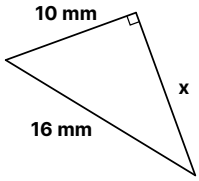
D Mixed practice ●●○

4. Find x. First decide whether x is the hypotenuse or a shorter side.

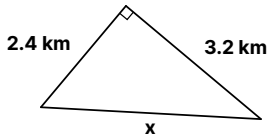
a)



b)



c)



E Is it a right-angled triangle? ●●○

Remember: It is right-angled only if the squares of the two shorter sides add up to the square of the longest side.

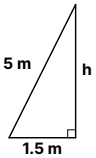
5. These are the side lengths of six triangles. Write yes if the triangle is right-angled and no if it is not.

a) 6 cm, 8 cm, 10 cm _____ **b)** 5 cm, 6 cm, 8 cm _____ **c)** 9 m, 40 m, 41 m _____

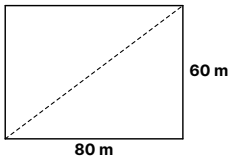
d) 7 m, 10 m, 12 m _____ **e)** 1.5 cm, 2 cm, 2.5 cm _____ **f)** 10 mm, 15 mm, 18 mm _____

F Problems ●●●

6. A 5 m ladder leans against a wall. The foot of the ladder is 1.5 m from the wall. How high up the wall does the ladder reach?



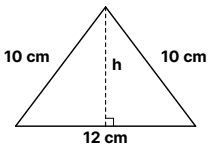
7. A rectangular field is 80 m long and 60 m wide. A path runs straight across it from one corner to the opposite corner. How long is the path?



8. A ship sails 12 km east and then 9 km north. How far is it from where it started?

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9. An isosceles triangle has two sides of 10 cm and a base of 12 cm. Find its height, h.



Extension

10. Find the distance between the points $(1, 2)$ and $(7, 10)$.

11. A box is 12 cm long, 4 cm wide and 3 cm high. Find the length of the longest straight rod that fits inside it.

12. The diagonal of a square is 10 cm long. Find the length of its sides.

13. A right-angled triangle has a hypotenuse of 25 cm and one side of 7 cm. Find its area.

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Pythagoras' Theorem

Name _____ Date _____ Calculator allowed Version 2

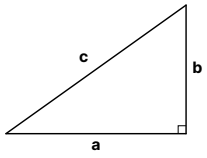
Scan to practise this topic online

KEY WORDS right-angled triangle *(tapatoru hāngai)* · hypotenuse *(tapa roa)* · square *(tapawhā rite)*

A The hypotenuse ●○○

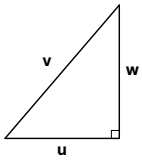
1. Write the letter of the hypotenuse of each triangle.

a)



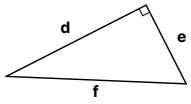
Hypotenuse: _____

b)



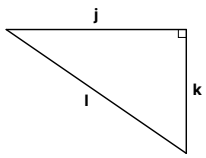
Hypotenuse: _____

c)



Hypotenuse: _____

d)



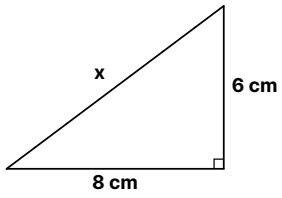
Hypotenuse: _____

B Finding the hypotenuse ●○○

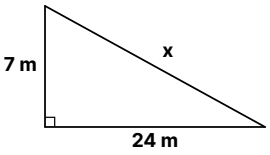
> **Remember:** $a^2 + b^2 = c^2$, where c is the hypotenuse, the longest side.

2. Find the hypotenuse, x. Round to 1 decimal place where needed. Diagrams are not to scale.

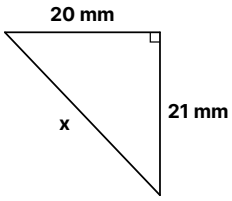
a)



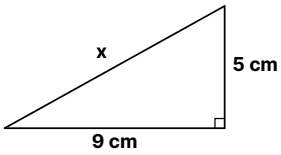
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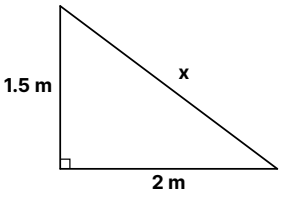
c)



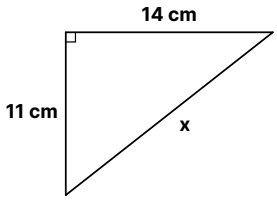
d)



e)



f)



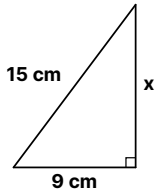
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C Finding a shorter side ●●○

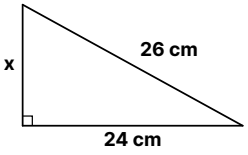
⚠ **Watch out:** Here the hypotenuse is given, so subtract: $a^2 = c^2 - b^2$.

3. Find the length of the side marked x. Round to 1 decimal place where needed.

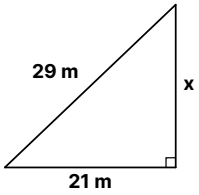
a)



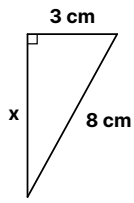
b)



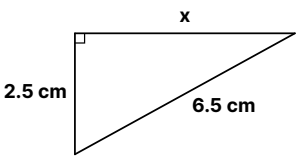
c)



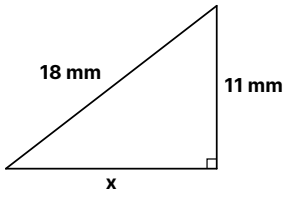
d)



e)



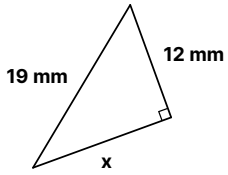
f)



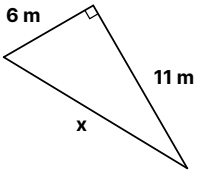
D Mixed practice ●●○

4. Find x. First decide whether x is the hypotenuse or a shorter side.

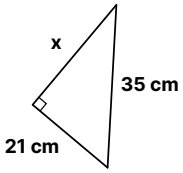
a)



b)



c)



E Is it a right-angled triangle? ●●○

Remember: It is right-angled only if the squares of the two shorter sides add up to the square of the longest side.

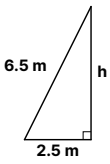
5. These are the side lengths of six triangles. Write yes if the triangle is right-angled and no if it is not.

a) 9 cm, 12 cm, 15 cm _____ **b)** 4 cm, 7 cm, 8 cm _____ **c)** 11 m, 60 m, 61 m _____

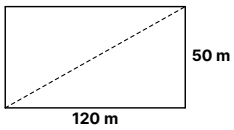
d) 8 m, 11 m, 14 m _____ **e)** 2.5 cm, 6 cm, 6.5 cm _____ **f)** 12 mm, 16 mm, 21 mm _____

F Problems ●●●

6. A 6.5 m ladder leans against a wall. The foot of the ladder is 2.5 m from the wall. How high up the wall does the ladder reach?



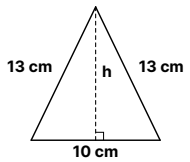
7. A rectangular field is 120 m long and 50 m wide. A path runs straight across it from one corner to the opposite corner. How long is the path?



8. A ship sails 15 km east and then 8 km north. How far is it from where it started?

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9. An isosceles triangle has two sides of 13 cm and a base of 10 cm. Find its height, h.



Extension

10. Find the distance between the points $(2, 1)$ and $(14, 6)$.

11. A box is 6 cm long, 3 cm wide and 2 cm high. Find the length of the longest straight rod that fits inside it.

12. The diagonal of a square is 8 cm long. Find the length of its sides.

13. A right-angled triangle has a hypotenuse of 17 cm and one side of 8 cm. Find its area.

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